When Labels Replace Mechanisms: Structural Bias in Identifying Susceptible Populations in Environmental Exposure Studies

Hanyu Wang

In research on environmental exposures and health effects, the identification of susceptible populations often remains confined to superficial demographic stratification, such as sex, age, race, or socioeconomic status. Analytical frameworks typically emphasize environmental exposure and SES variables, while overlooking other factors that may drive observed differences between groups. This simplified approach may overlook the key mechanisms that cause group differences, thereby leading to misleading inferences about the causes of health disparities, and conflating structural exposure differences with biological susceptibility. The impact of air pollution on coronary heart disease provides a pertinent example.

Many studies have confirmed a clear association between air pollution and cardiovascular diseases (Lee et al., 2014). When discussing the impact of wildfire smoke or PM2.5 on coronary heart disease, most studies analyze susceptible populations, and findings suggest that risk is higher in women than in men. However, such studies often fail to fully consider the differences in the actual exposure structure between men and women.

For instance, Baumgartner et al. (2014) discovered in rural areas of China that black carbon exposure was significantly associated with elevated blood pressure in women, and this correlation was more pronounced among women living near highways. This may reflect the fact that rural women are more frequently involved in charcoal-burning or garbage-burning activities, while the regulation of burning near highways is weaker, resulting in higher exposure intensity. On the contrary, Weichenthal et al. (2014) found in rural areas of Iowa and North Carolina in the United States that PM2.5 was associated with cardiovascular mortality in men, but no similar association was observed in women. This difference may be due to the fact that local men are more engaged in long-term outdoor agricultural labor, inhaling more diesel exhaust.

These two studies suggest that the gender differences may stem from the differential exposure resulting from the division of social and family roles, rather than purely biological differences. Without explicit modeling of differential exposure intensity and role-based environmental contact patterns, observed sex differences may represent exposure heterogeneity rather than true biological effect modification. This leads to a hypothesis that is worthy of further examination: the findings that women have a higher risk of coronary heart disease due to air pollution, might partly be because women undertake more cooking tasks and inhale more kitchen fumes, rather than reflecting intrinsic biological vulnerability. However, most current studies do not incorporate social role-related work and family exposures into the analytical framework, resulting in gender differences often being assumed to be differences in physiological susceptibility between men and women. This failure to decompose structural and biological components of risk may obscure the true social and behavioral mechanisms driving the observed health disparities between groups.

References

Baumgartner, J., Zhang, Y., Schauer, J., Wang, Y., Huang, W., & Ezzati, M. (2014). Highway proximity and Black Carbon from cookstoves as a risk factor for higher blood pressure in rural China. *ISEE Conference Abstracts*, 2014(1). https://doi.org/10.1289/isee.2014.o-185

Lee, B.-J., Kim, B., & Lee, K. (2014). Air Pollution Exposure and cardiovascular disease. *Toxicological Research*, 30(2), 71–75. https://doi.org/10.5487/tr.2014.30.2.071

Weichenthal, S., Villeneuve, P. J., Burnett, R. T., van Donkelaar, A., Martin, R. V., Jones, R. R., DellaValle, C. T., Sandler, D. P., Ward, M. H., & Hoppin, J. A. (2014). Long-term exposure to fine particulate matter: Association with Nonaccidental and cardiovascular mortality in the Agricultural Health Study Cohort. *Environmental Health Perspectives*, 122(6), 609–615. https://doi.org/10.1289/ehp.1307277